

MICA — evidence figures

Evidence provenance: hand-rule, observation-learned, or both

Every number in this document was produced by one of three kinds of machinery, and each figure below carries its tag. **Hand-rule:** deterministic, hand-coded computation (the template matcher, the safety gate’s rules, the Bayesian filter’s update equations). **Observation:** components learned from watching — the frozen foundation encoders (VPT, MineCLIP, Uni3D), the likelihood heads trained on recorded sessions, the action decoder trained on the human’s actual next moves. **Both:** hand-specified structure carrying learned content.

Figure / finding	hand-rule part	observation part	type
Fig. 1 necessity	the reactive baseline itself; filter equations	likelihood heads over frozen encoder features	both
Fig. 2 pretraining	target-building rules only (mistake pairing, hygiene)	learning signal: the human’s observed next actions	obs.
Fig. 3 contested	matcher groups the sessions	learned heads; builder’s label as supervision	both
Fig. 4 convergence	coverage bookkeeping	the frozen shape embedding	obs.
Fig. 5 case study	filter dynamics	learned likelihoods	both
Fig. 6 anticipation	the τ -match metric	the decoder, end to end	obs.
matcher 1.0 + boundary	entirely hand-rule	—	hand
gate / consent / A7	entirely hand-rule (by design)	—	hand

The separation is itself a finding. The **hand-rule components are where trust lives**: exact on known truth (matcher 1.0 on the scripted corpus), inspectable, and identical across experiment arms — but they collapse off-template (0.042, Fig. 1), and that collapse is measured, not assumed. The **observation-learned components are where generalization lives**: they survive off-template (0.375), close domain gaps (Fig. 2), and improve lawfully with more observation (Figs. 2–4). MICA’s architecture assigns roles accordingly — *hand rules govern safety and judgment* (the gate, consent, the do-operator), *learned components govern perception and anticipation* — and Fig. 1 is the head-to-head justification for that division: the hand-rule system alone dies exactly where real work lives, while the hybrid keeps working.

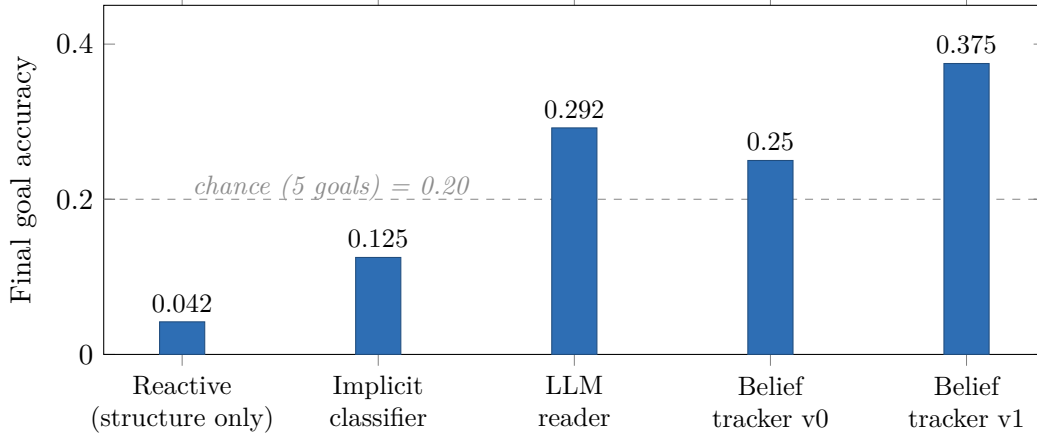


Figure 1: **Intent inference is necessary off-template.** Final goal accuracy on 24 real free-build sessions that no arm ever trained on. The reactive baseline—structure matching alone, the “helper without intent inference”—collapses to 0.042, while the belief tracker (v1) reaches 0.375, a $9\times$ gap. Exactly where builds stop matching templates, structure alone stops working. (Four-arm comparison, exposure-verified splits, cascade of 2026-07-19.) *Evidence type: BOTH* — the baseline is the hand-rule system itself; the tracker is learned likelihoods inside a hand-specified filter.

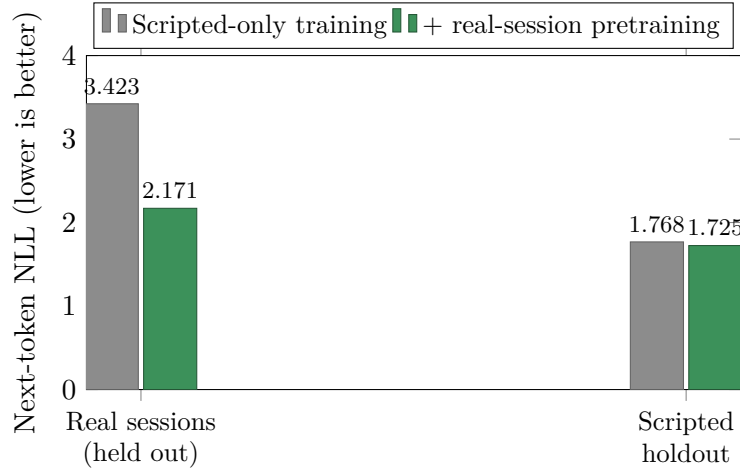


Figure 2: **The decoder’s domain gap, and closing it without labels.** The action decoder trained only on scripted builds is 1.25 nats worse at predicting a real human’s next building action than a scripted one (gray). Pretraining its next-token stage on real sessions’ action streams—usable from *every* session, contested or discarded, because no goal label enters the sample—recovers most of the gap (-37% on held-out real sessions) while slightly improving the scripted holdout. Pre-registered comparison, holdout sessions fixed by id before the run. *Evidence type: OBSERVATION* — the learning signal is the human’s actual next actions; hand rules only shape the targets.

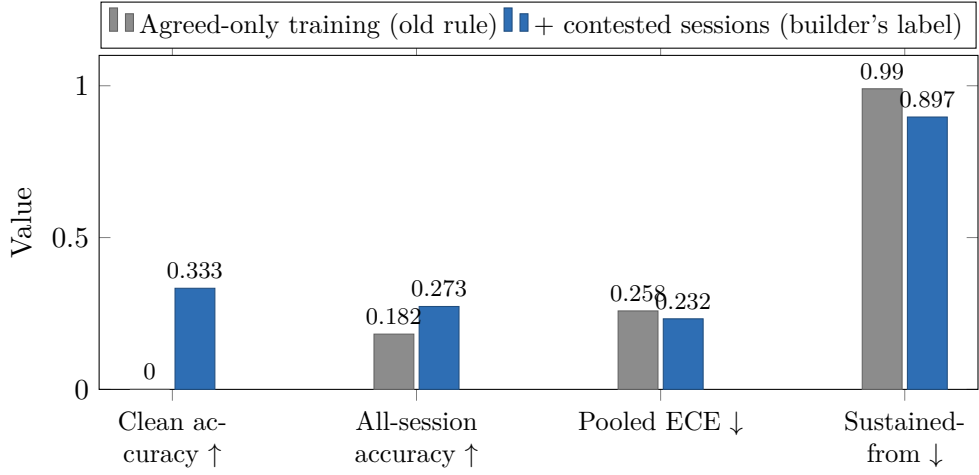


Figure 3: **The builder’s word is good training data.** Training the likelihood heads on matcher-contested sessions—with the builder’s own label as truth—improves every pre-registered criterion on a shared evaluation set neither variant trained on: accuracy on both session groups, calibration (ECE), and *earliness* (sustained-from: the build-progress fraction from which the top goal stays correct; lower = earlier). An identical experiment on 2026-07-12 found the opposite; the reversal followed the repair of a feature leak and the evidence corpus, and is monitored at every retrain. Caveat: the evaluation set is small (11 sessions). *Evidence type: BOTH — learned heads over observed features; the builder’s own label as supervision; the hand-rule matcher only groups sessions.*

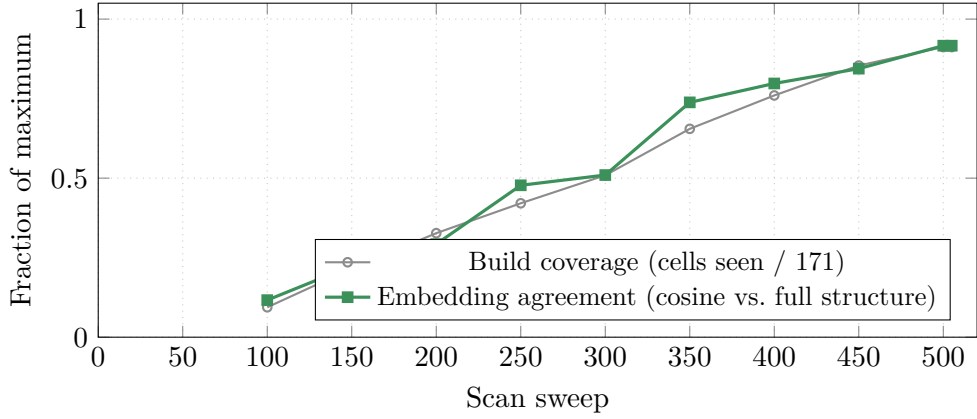


Figure 4: **Structural understanding accumulates with observation.** As the embodied agent’s first-person raycast scan covers more of the human’s build (gray), the 3D shape embedding computed from *only what the agent has seen* converges to the embedding of the complete structure (green, cosine similarity 0.92 at 91% coverage). The agent’s partial view carries the same structural signal as the full ground truth. (Session *fabric-20260719-052914*.) *Evidence type: OBSERVATION — the claim is about the frozen learned embedding; only the coverage count is hand bookkeeping.*

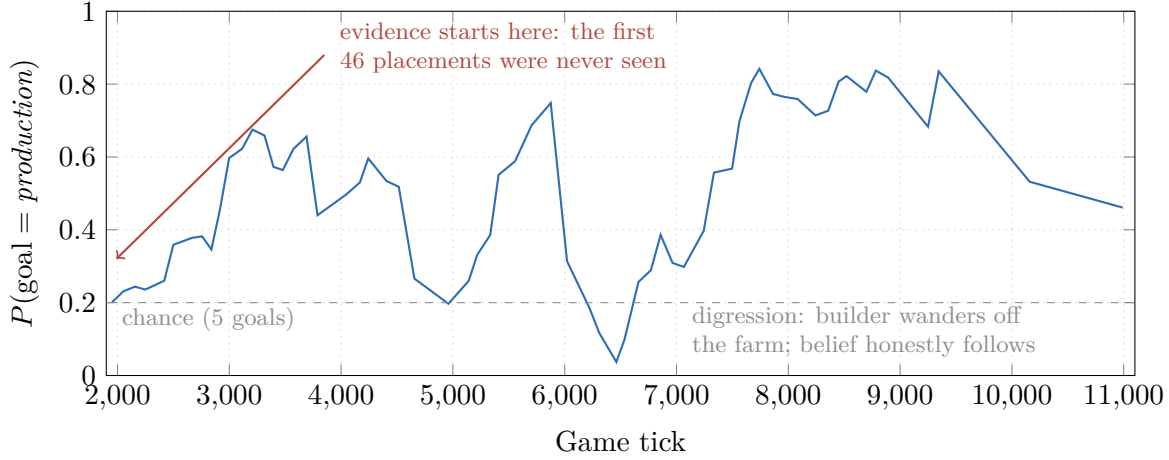


Figure 5: **Case study: tracking a real intention through evidence, absence, and digression.** A live session in which the builder made a crop farm (their own post-session label; goal category *production*). The belief starts at chance because the live pipeline attached late and missed the first 46 placements—yet recovers the correct goal from the remaining evidence, dips when the builder wanders mid-session (an honest response to contradicting evidence, not noise), and re-locks as the farm takes shape, peaking at $P = 0.84$. Illustrative single-session view; quantitative claims rest on the pooled evaluations of Fig. 1. *Evidence type: BOTH* — *hand-specified filter dynamics carrying learned likelihoods over live observed evidence.*

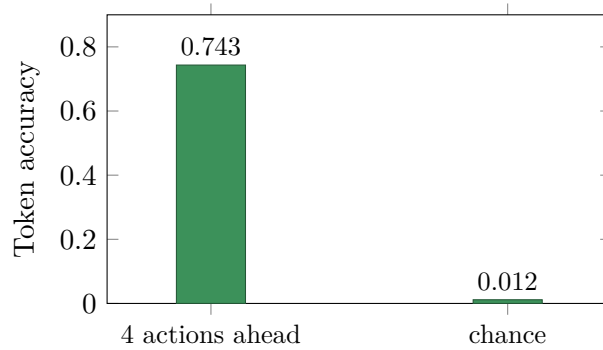


Figure 6: **Anticipation, not just recognition.** The multi-token decoder predicts the builder’s action *four steps ahead* with 74% token accuracy on held-out sessions, against a token chance level of 1.2% (87-token vocabulary, τ -matching: exact for symbolic tokens, ± 2 for coordinates). Understanding an intention here means correctly continuing the build several actions into the future. *Evidence type: OBSERVATION* — *the decoder is learned end to end; only the τ -match metric is hand-defined.*